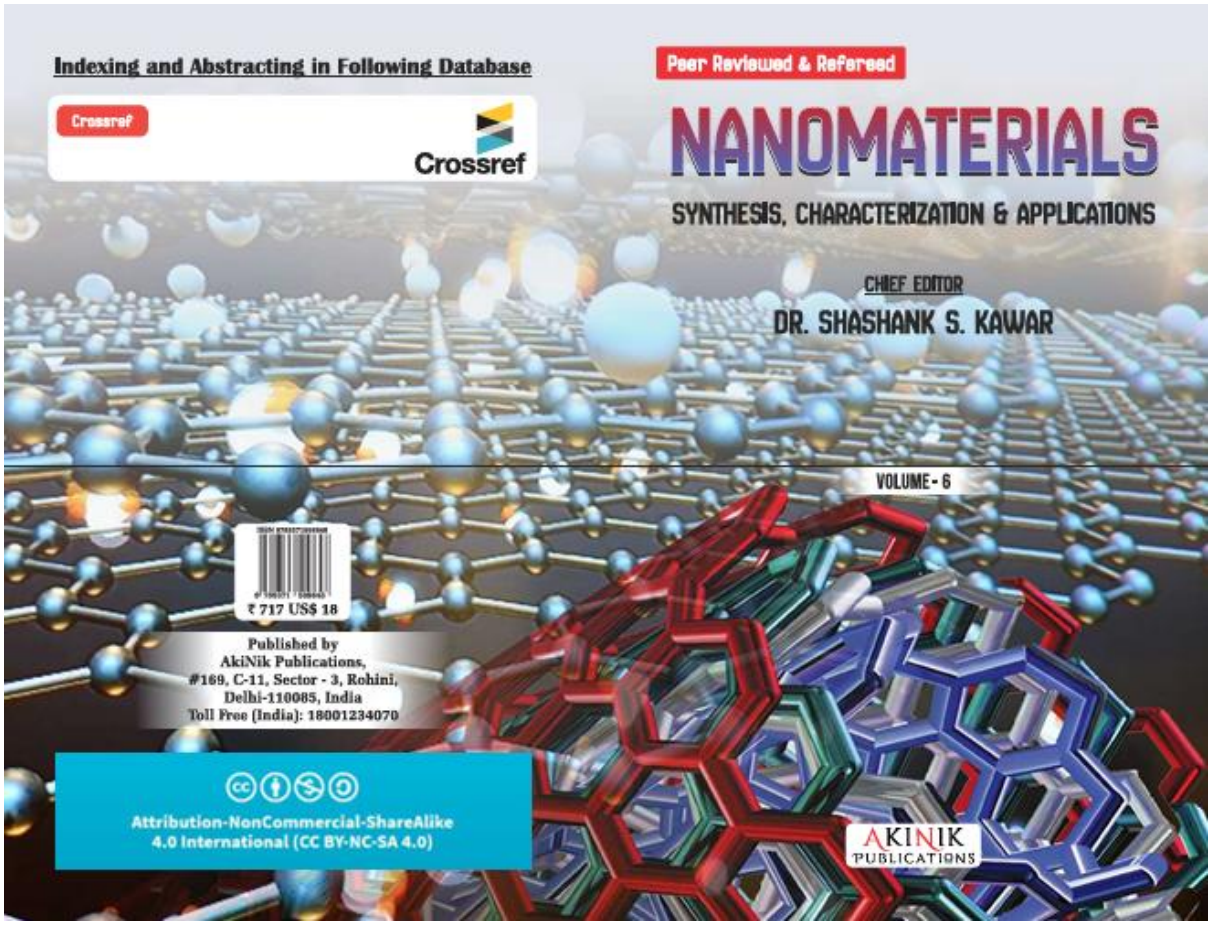
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Chapter - 8
Bio-functional Cellulose-based Thin Films for Wound Healing: Fabrication, Characterization and Biomedical Applications

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Chapter - 8

Bio-functional Cellulose-based Thin Films for Wound Healing: Fabrication, Characterization and Biomedical Applications

Venkatesan R, Varshini J, Suganya S and Ithayapriya B

Abstract

Wound healing is a complex biological process that requires advanced biomaterials capable of providing mechanical protection, maintaining moisture balance, preventing microbial infection, and supporting tissue regeneration. Conventional wound dressings mainly act as passive protective barriers and often lack therapeutic functionality, creating the need for multifunctional materials with enhanced healing capabilities. In recent years, cellulose-based biomaterials have attracted considerable attention in biomedical applications due to their excellent biocompatibility, biodegradability, hydrophilic nature, film-forming ability, and sustainable origin.

This chapter provides an overview of biofunctional cellulose-based thin films for wound healing applications, focusing on their fabrication methods, characterization techniques, and biomedical potential. The structural characteristics of cellulose and its derivatives, along with their suitability for developing advanced biomaterial platforms, are discussed. Various fabrication approaches, including solvent casting, dip coating, and spin coating, are reviewed, with particular emphasis on spin coating for producing uniform films with controlled thickness and improved surface properties.

The incorporation of bioactive molecules into cellulose matrices enhances their therapeutic performance. Specifically, salicylic acid-loaded cellulose thin films are highlighted as promising wound dressing systems due to their antimicrobial and anti-inflammatory properties. Key characterization aspects, including film thickness evaluation, mechanical strength analysis, and antimicrobial assessment, are presented to understand the functional performance of these materials.